Development Workflow: Sovereign Web Application

Project Scope: Zero-Budget Scalable Architecture | Tech Stack: React, R3F, Firebase, Vercel

Phase 1: Architecture & Foundation (Weeks 1-2)

1.1 Environment & CI/CD

Initialize Vite + React project. Connect repository to Vercel for automated, free-tier deployments. Establish the production pipeline to ensure zero-lag delivery via global CDNs.

1.2 Firebase Backend Integration

Configure Firebase Authentication. Design Firestore NoSQL schemas for **Users**, **Planets**, **Trackers**, and **Clans**. Implement Firebase Realtime Database specifically for **Fchat** to handle high-frequency messaging without exceeding free read/write limits.

Phase 2: Onboarding & Visual Sequences (Weeks 3-4)

2.1 Cinematic Intro & Transitions

Embed pre-rendered, highly optimized MP4 videos for the intro sequence and wormhole travel. Using static video instead of real-time 3D particle generation guarantees zero lag and prevents browser crashing on lower-end devices.

2.2 Dynamic Solar System Selection

Integrate React Three Fiber (R3F). Render basic spherical meshes for planets. Build an admin control panel allowing you (the owner) to write a single integer to Firebase (e.g., `total_planets = 5`), which dynamically updates the number of selectable planets for new users.

Phase 3: Core Loop - MyTube & AI Trackers (Weeks 5-7)

3.1 MyTube Interface

Build the video tutorial interface using Tailwind CSS, mimicking standard video platforms. Host tutorial videos externally (YouTube unlisted) and embed them via iframes to save bandwidth.

3.2 AI Tracker Generation

Integrate a free-tier AI API (e.g., Google Gemini). Build a sliding sidebar UI. The user inputs their goal, and the AI outputs a structured JSON response that your app parses to automatically build and save their custom tracker to Firestore.

Phase 4: Progression & Mechanics (Weeks 8-10)

4.1 Mathematical Point Uniformity

Establish the universal formula linking tracker consistency to planetary development points.

```
DevPoints = (BaseTaskValue × StreakMultiplier) + EventBonus
```

4.2 Planet & Moon Evolution

Map DevPoints to 3D asset visibility in R3F. Level 0 planets render as barren spheres. As DevPoints reach predefined thresholds, script the automatic un-hiding of 3D buildings, atmospheric effects, and user-developable moons.

Phase 5: Gameplay - Combat & Clan Wars (Weeks 11-13)

5.1 Asteroid Mining & Combat Logic

Implement a mathematically balanced PvE combat system for asteroid mining. Compare the user's tracker-earned power against the asteroid creature's static defense rating.

```
WinProbability(%) = [ (UserDevPoints / 100) / CreatureDefense ] × 100 ± RNG(10)
```

5.2 Fchat & Clan Wars

Finalize the global chat interface. Build the Clan War system where weekly point deltas from user trackers are aggregated. The clan with the highest growth metric wins bonus DevPoints and unique planetary aesthetic upgrades.

Phase 6: 3D Assets & Polish (Weeks 14-16)

6.1 Optimized Asset Loading

Source free 3D models (Armies, Dragons, Civilians, Creatures) from platforms like Mixamo. Compress all files using Draco compression (.gltf / .glb) to reduce file sizes by up to 80%, ensuring the app never hangs during loading screens.

6.2 Final QA & Event System

Implement the Live Event architecture (a central Firebase document that pushes limited-time multipliers or boss asteroids to all users simultaneously). Conduct performance audits using Chrome DevTools.

Phase 7: Security & Vercel Launch (Weeks 17-18)

7.1 Firebase Security Rules

Lock down the database. Write strict Firestore rules so users can only edit their own trackers and profiles. Ensure only the admin (you) can alter global events, creature stats, or the solar system's planet count.

7.2 Production Build & Deployment

Run the final Vite production build. Connect your custom domain to Vercel. Perform a Lighthouse audit to ensure the site hits a 90+ performance score, confirming the zero-lag requirement is met before the public sees it.

Phase 8: Live Operations & Scaling (Weeks 19+)

8.1 First Live Event & Analytics

Trigger the first global event (e.g., "Asteroid Swarm") by updating your central Firebase config. Monitor traffic and performance completely free using Vercel Analytics and Firebase Crashlytics.

8.2 Clan War Resolution & Fchat Moderation

Establish a weekly routine to reset Clan War scores. Distribute the DevPoint bonuses to the winning clan and automatically deploy their cosmetic 3D planet upgrades. Actively monitor Fchat to maintain a healthy community.